

Math 141

Final Exam Review Sheet

This review sheet is meant to help you self-assess your understanding of course material that has been covered *since the midterm*. You should also look back over the midterm review sheet, though the majority of the exam will focus on topics we've covered since the midterm.

The questions provided here are not all representative of how the exam questions will be asked, and the final will assess several, but not all, of these concepts. Use your lecture notes, completed assignments, and the posted slides as references while you work through the review sheet. Feeling comfortable answering the conceptual questions and being able to complete the selected practice problems, while timing yourself and without referencing your notes, are very good indicators that you are ready for the exam. You will not be asked to code on the exam, but may be asked to write pseudo-code (which we will practice in lab) or to explain code that is provided to you.

You are allowed two 3x5 index cards with handwritten notes on both sides, but otherwise, you are not allowed to use *any* notes, textbooks, or any other kind of resources on the exam. I will include theory-based inference summary tables in the exam.

NOTE: You will be required to submit your note cards along with your final exam.

Remember that office hours (both with me and the course assistants), the Slack channel, and collaboration with your peers are available to you as resources while you prepare for the exam.

Inference overview

Suppose we want to know how many *more* gallons of kombucha the average Portlander consumes in a year compared to the average Boulderite (from Boulder CO). We do a survey of 100 randomly sampled Portlanders and 100 randomly sampled Boulderites and ask them about their yearly intake.

- What is the population, the sample, the parameter, and the statistic of interest?
- Explain the differences and relationships between the following distributions and explain why we care about them for inference:
 - i. Population distributions of annual kombucha intake in each city
 - ii. Sampling distribution of the statistic
 - iii. Bootstrap distribution of the statistic
 - iv. Null distribution of the statistic (assuming our null hypothesis is that the parameter = 0).
- Suppose the value of our statistic is 1.5, and that the true parameter value is 2. What is the center of the distributions from ii, iii, and iv above?
- What kind of evidence towards our research question do confidence intervals provide us with? What about hypothesis tests? Which distributions do we use to find confidence intervals vs hypothesis tests?
- We've discussed theory-based inference for four specific summary statistics: a proportion, a difference in proportions, a mean, and a difference in means. When do we use each of these? Consider the following possible data settings:
 - One categorical variable (ex. `var1` = whether or not each Portlander in our sample drinks any kombucha)
 - Two categorical variables (ex. `var1` = whether or not each person in our sample drinks any kombucha, and `var2` = whether they live in Portland or Boulder)
 - One quantitative variable (ex. `var1` = how many gallons of kombucha each Portlander in our sample drinks)
 - One quantitative variable and one categorical variable (ex. `var1` = how many gallons of kombucha each person in our sample drinks, and `var2` = whether they live in Portland or Boulder)
- Revisit the conditions for inference for each of these (you'll also be able to find these in the theory-based inference summary table I provide on the exam)

Confidence intervals

- Explain the difference between the percentile and SE methods for constructing confidence intervals, including how we generally calculate each and conditions for when we should use each method.
- How do confidence intervals relate conceptually to sampling distributions? (*Hint: review slides from 3/9*)
- How do confidence intervals change if: we change the sample size, we change the confidence level, or we take a new sample from the population?
- Suppose we are given a 90% confidence interval of $[1.5, 3]$ for some statistic. How should we interpret this confidence interval? Why would it be wrong to say “there is a 90% chance the parameter is in our interval?”
- What are the general steps we follow to construct simulation-based confidence intervals?
- What are the general steps we follow to construct theory-based confidence intervals? For the SE method, what are “critical values” and how do we find them?
- Some assignment/lecture exercises to review (*Note: you won't be asked to use R, but will be expected to know where relevant values come from conceptually, ex. what quantiles you need from what distribution*):
 - HW 6 Exercise 1 (simulation-based percentile method confidence intervals)
 - HW 6 Exercises 4 & 5 (simulation-based percentile & SE method confidence interval)
 - Lab 9 Exercise 6b & HW 9 Exercise 1g (theory-based confidence intervals for mean, difference in means)
 - Right side of the activity from the end of 4/6 slides (theory-based confidence intervals for a proportion)

Hypothesis testing

- What are the general steps for the hypothesis testing framework?
- What is the general interpretation of a p-value?
- What does it mean if we set $\alpha = 0.10$? What are the effects of changing α ?
- What are Type I and Type II errors? How do we control each of these?
- What is power, and why is it important?
- What distinguishes a 1-sided test from a 2-sided test? Draw pictures of the null distribution and what parts of it represent the p-value for each type of test.
- What are the general steps we follow to perform a simulation-based hypothesis test?
- What are the general steps we follow to perform a theory-based hypothesis test?
- Some assignment/lecture exercises to review (*Note: you won't be asked to use R, but will be expected to know where relevant values come from conceptually, ex. the distribution and probability needed to calculate a p-value*):
 - ESP example from slides on 3/16 (simulation-based hypothesis testing)
 - HW 7 Exercise 1 (hypothesis testing framework and errors)
 - Lab 7 Exercise 3 (simulation-based hypothesis testing)
 - Left side of the activity from the end of 4/6 slides (theory-based hypothesis testing for a proportion)
 - Lab 9 Exercise 5 (theory-based hypothesis testing for a difference in proportions)
 - HW 9 Exercise 1 b-f (theory-based hypothesis testing for a difference in means)

Random variables

- How do we define and calculate expected value of a random variable?
- How do we define and calculate variance of a random variable?
- When X and Y are two random variables and c is a fixed number, how/under what conditions can we calculate:
 - $E[X + Y]$
 - $E[cX]$
 - $Var[X + Y]$
 - $Var[cX]$
- What is an example of a Binomial distributed random variable? What are the relevant parameters? What are the formulas for the mean and variance of a Binomial random variable?
- What is an example of a Normally distributed random variable? What are the relevant parameters?
- What do we mean by the “Standard Normal” distribution? What does standardizing do?
- In your own words, what does the Central Limit Theorem tell us, and why is it important for statistical inference?
- Practice problems:
 - HW 8: Exercises 1 and 5
 - Activity from the end of the 3/30 slides
 - Activity from the end of the 4/1 slides

Inference for regression

- Be sure to review the midterm study guide, particularly on interpretation of coefficients and writing equations for the line of best fit (aka the fitted model, the estimated model).
- How can we perform theory-based hypothesis tests and obtain confidence intervals using `get_regression_table()`?
- What are the default null and alternative hypotheses for regression coefficients from `get_regression_table()`?
- What is the default confidence level for confidence intervals from `get_regression_table()`?
- What are the conditions for theory-based inference for regression?
 - How do we check those conditions?
 - Draw a picture or give an example of when each condition is met vs not met.
- When should or can we do simulation-based inference for regression instead?
- What are the general steps for doing simulation-based inference for regression (both confidence intervals and hypothesis tests)?
- Should p-values be used for model selection? What are other model-selection approaches we can use?
- What hypotheses does ANOVA test? Generally speaking, how does it compare our observed data to the data we'd expect to see under the null?
- For practice:
 - Review lab 10

Data wrangling, visualizing, and resampling in R

- What do each of the following `dplyr` functions do? Revisit slides from 2/4 for examples.
 - `count()`
 - `filter()`
 - `arrange()`
 - `mutate()`
 - `case_when()`
 - `summarize()`
 - `group_by()`
- Practice explaining code that is shown to you:
 - HW 2 Exercise 3
 - Q3 of the in-class Sampling vs Bootstrap Distributions activity
- Practice writing pseudo-code for data wrangling tasks:
 - Exercise at the end of the 2/4 slides
 - Try the same type of exercise with the instructions for HW 2 Exercise 5a and 5b
- Review the components of the grammar of graphics (slides from 1/28 and 1/30)
- Revisit your work on lab 2. How should arguments to `aes()` change depending on the graph you are creating? How should `geom_----()` change according to the graph you are creating?
- Suppose we have data on our sample of kombucha-drinking Portlanders and Boulderites. Each row of our data frame represents one respondent. The columns of our data frame are `id` to distinguish each individual surveyed, `city` for whether they live in Portland or Boulder, `kombucha` which measures how much kombucha they drink annually, and `indie_folk` which indicates how many hours of indie-folk music they listen to per year. Suppose we want to make a scatterplot of `kombucha` as our response variable and `indie_folk` as our explanatory variable, and color the points by `city`.
 - What should we set `geom_----` to be?
 - What should we put inside `aes()`?